Horizontal and vertical dilations

1

a 360

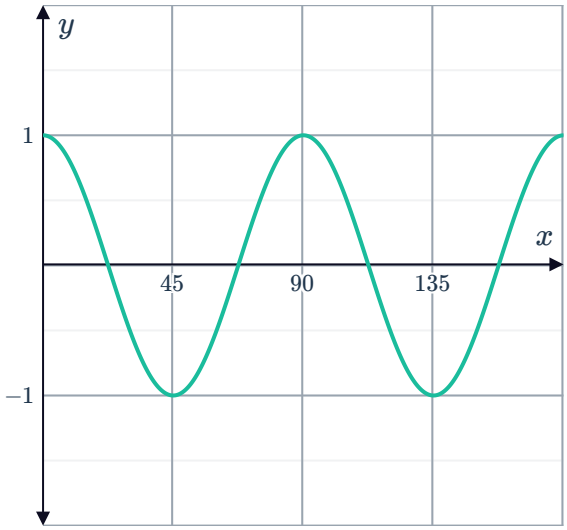
b

x	0	22.5	45	67.5	90	112.5	135	157.5	180
$g(x)$	1	0	-1	0	1	0	-1	0	1

c 90

d Horizontal dilation by a factor of $\frac{1}{4}$

e

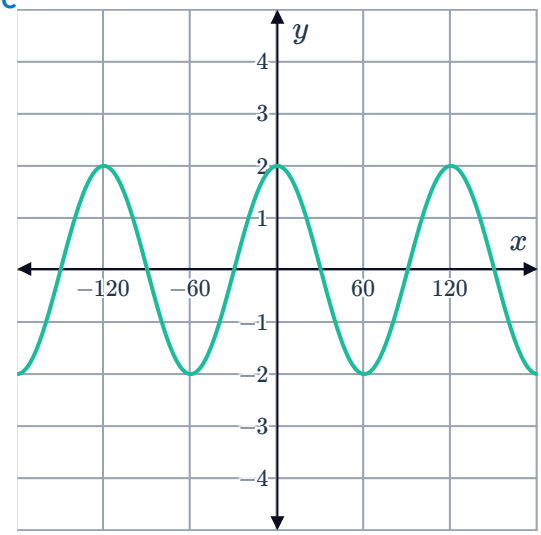


2

a 2

b 120

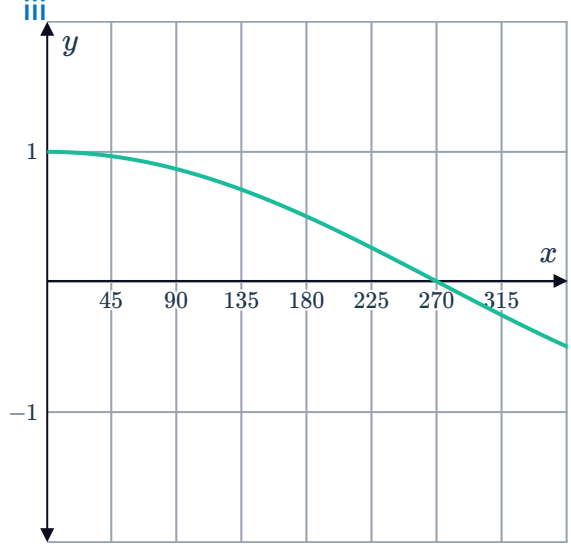
c



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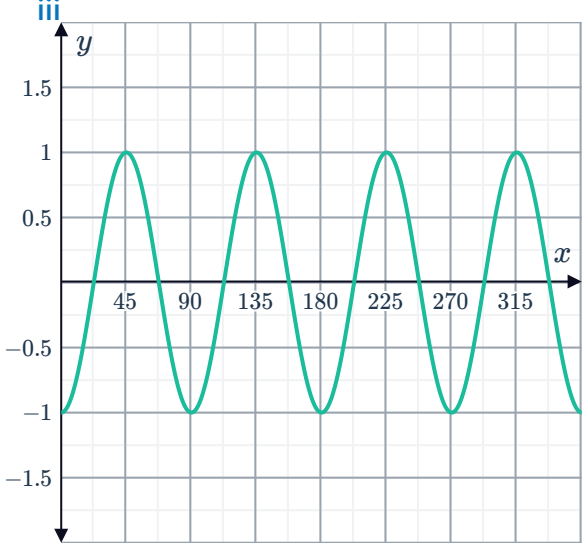
d

i $A = 1$ ii $P = 1080$ iii



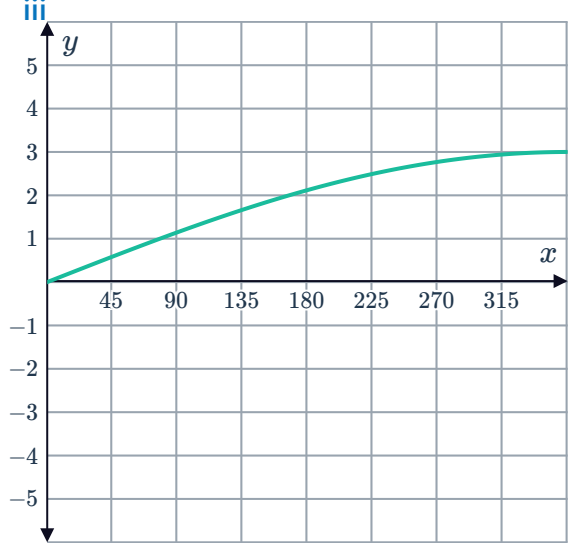
e

i $A = 1$ ii $P = 90$ iii



f

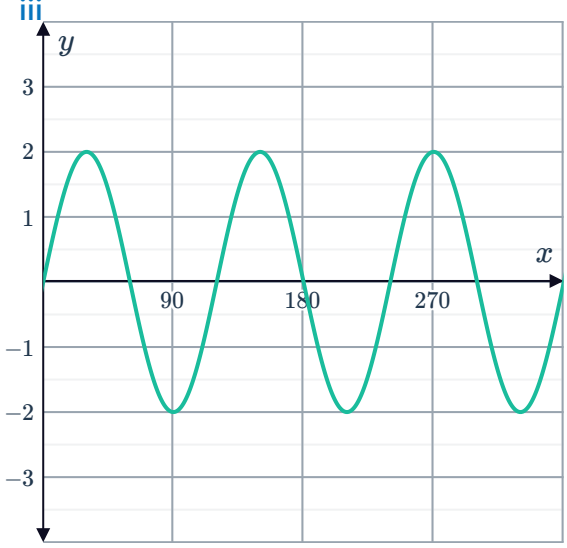
i $A = 3$ ii $P = 1440$ iii



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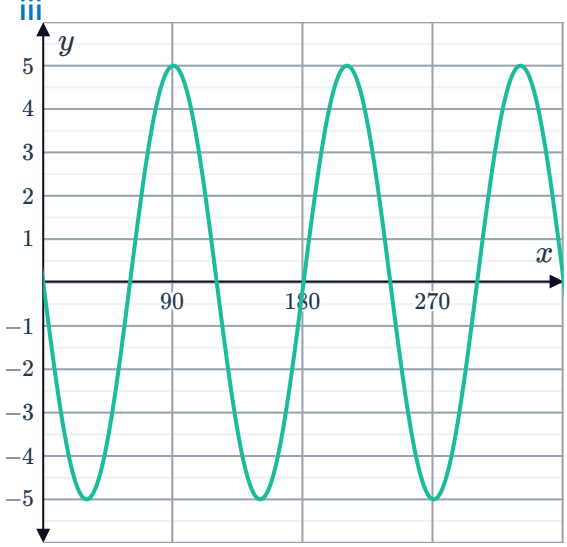
g

i $A = 2$ ii $P = 120$ iii



h

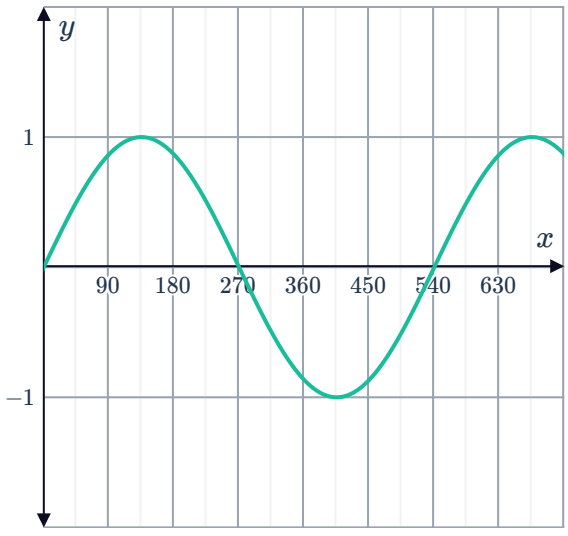
i $A = 5$ ii $P = 120$ iii



4

a 1 **b** 540°

c



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5

a

x	0	360	720	1080	1440
$\sin\left(\frac{x}{4}\right)$	0	1	0	-1	0

b 1440

6

a Horizontal dilation by a factor of $\frac{1}{4}$.

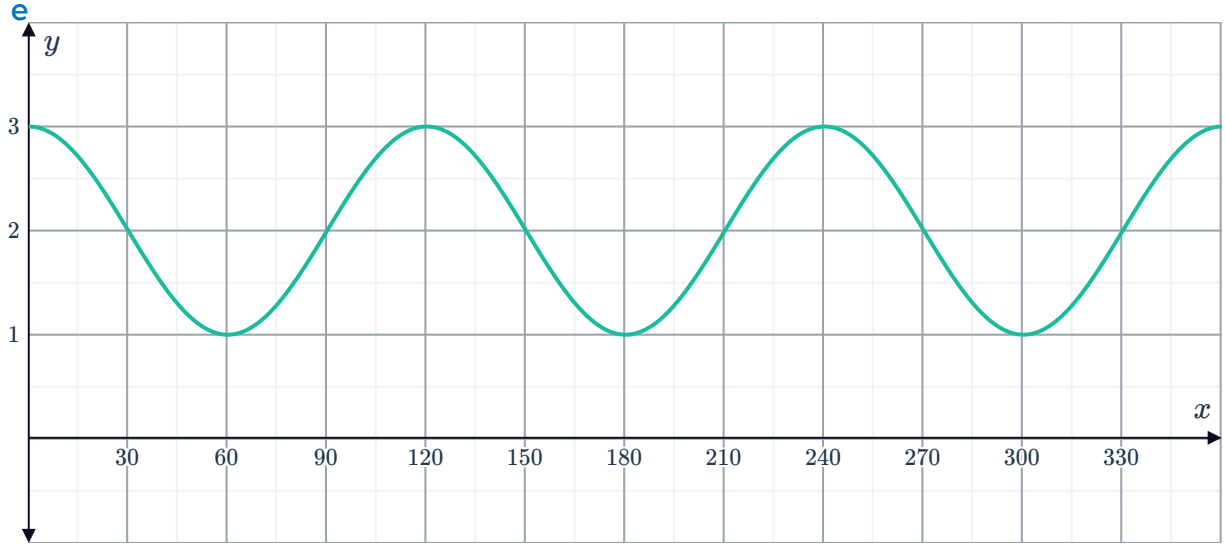
b 4

Combined translations and dilations

7

a 120 **b** 1 **c** 3 **d** 1

e



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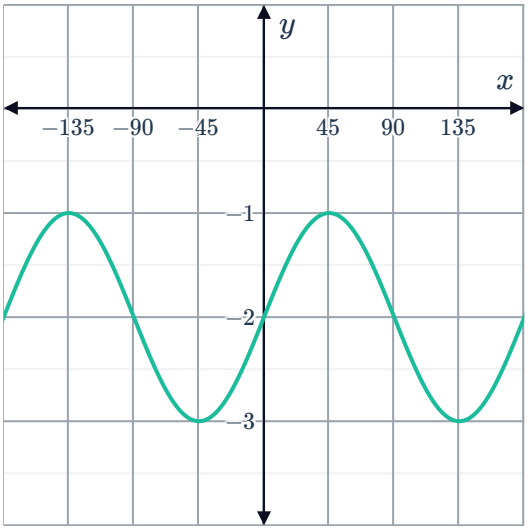
8

a

i $(-\infty, \infty)$

ii $[-3, -1]$

iii

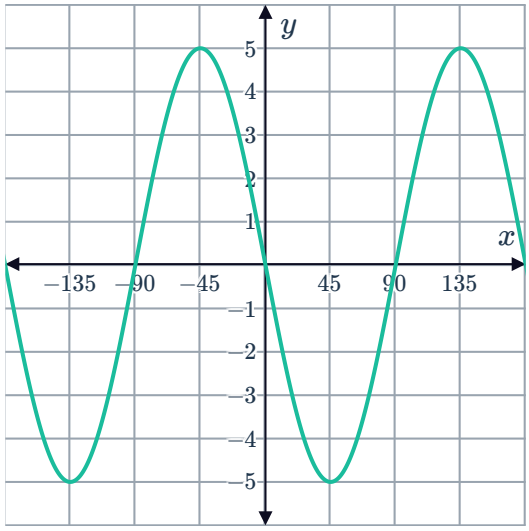


b

i $(-\infty, \infty)$

ii $[-5, 5]$

iii

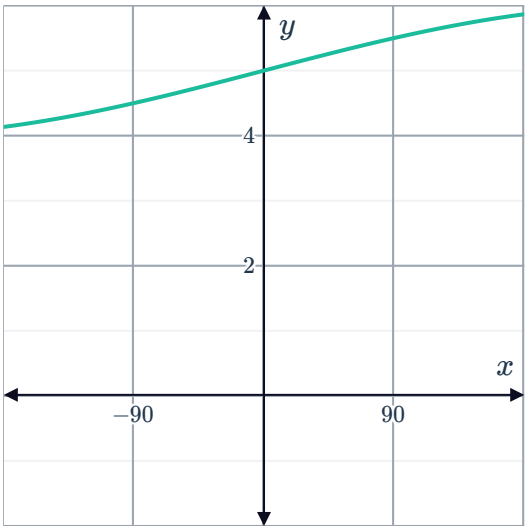


c

i $(-\infty, \infty)$

ii $[4, 6]$

iii

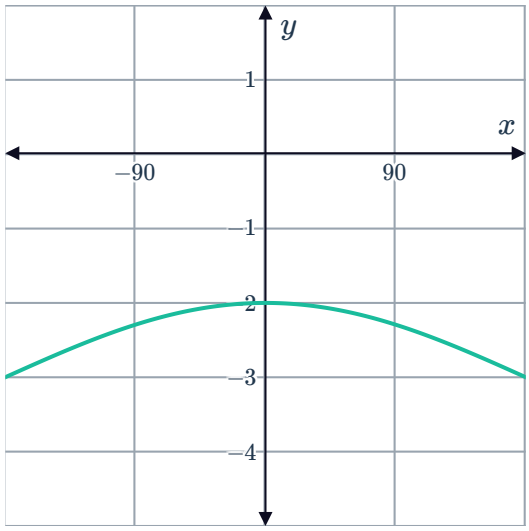


d

i $(-\infty, \infty)$

ii $[-4, -2]$

iii



9

a $y = 3\sin 2x$ **b** $y = -\sin 2x$ **c** $y = 2\cos 3x$ **d** $y = -4\cos\left(\frac{x}{2}\right)$

10

a Yes **b** No **c** Yes **d** No

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